Conspiracy language markers and detection

April 13, 2026

Research Goal: I want a wide academic literature search for a research project proposal on conspiracy theories in online text, centered on conspiracy-language detection and psycholinguistically grounded marker extraction rather than multilingual transfer. The focus should include work on identifying and localizing linguistic markers of conspiratorial thinking in raw text, such as us-versus-them framing, appeals to hidden knowledge, attribution of malicious intent to powerful groups, and epistemic mistrust, as well as document-level detection that distinguishes endorsement of conspiracy theories from merely reporting on them. The literature should be mixed rather than purely computational: include NLP papers on conspiracy detection, span annotation/extraction, explainable or interpretable modeling, related work from misinformation and propaganda span detection when relevant, and theory-oriented work from psycholinguistics, cognitive psychology, and social psychology that motivates these markers. The search should be useful for building a proposal and should prioritize papers relevant to datasets with raw text, for example datasets such as LOCO. We're also interested in adding conspiracy theories to the pretraining set.

Found 51 papers · April 13, 2026 · Estimated coverage of relevant papers: 97%

Summary of Results

The clearest pattern is that useful conspiracy-language modeling now depends less on topic words alone and more on separating conspiratorial endorsement from adjacent but non-conspiratorial opposition, then localizing the rhetoric that carries that distinction [1], [2], [3].

Detection targets and datasets

- **LOCO** is the main raw-text resource for topic-matched conspiracy vs. mainstream documents at scale, with broad linguistic metadata and a prototypical conspiracy subset; it is well suited for feature discovery and domain-adaptive pretraining, but its labels are document-level and do not by themselves resolve endorsement vs. reportage [4], [5].
- **Stance-aware Twitter datasets** explicitly distinguish support, neutrality, and opposition to conspiracy claims, which matters for avoiding false positives on discussion or debunking [6], [7], [8].
- **Oppositional Thinking / XAI-DisInfodemics** is the strongest match for marker extraction: it pairs document classification with token-level spans for oppositional narrative elements and directly contrasts conspiracy with merely critical discourse [1], [2].

Marker families that recur

- **Intergroup conflict / us-versus-them** and violence/anger cues are strong discriminators [1], [9].
- **Hidden knowledge, alternative-truth framing, evidentiality/epistemic modality, and institutional abuse metaphors** emerge in corpus analysis [10], [11].
- **Malicious powerful agents, actions, and targets** appear as reusable narrative motifs across topics [12], [13].
- **Generalized mistrust and rejection of official explanations** are the main psycholinguistic anchors from theory [14], [15], [16].

A proposal centered on interpretable marker extraction is therefore best framed around multi-granular supervision—document label + stance + spans/rationales—because annotation is genuinely ambiguous and hard-label setups obscure the phenomenon [17], [18], [19].

Paper Catalog (51 papers)

	Year	Cit/yr	Title	Authors	Journal
1	2024	3.4	What distinguishes conspiracy from critical narratives? A computational analysis of oppositional discourse (link)	Damir Korenčić et al.	Expert Systems
2	2024	8.0	Overview of the Oppositional Thinking Analysis PAN Task at CLEF 2024 (link)	Damir Korenčić et al.	Conference and Labs of the Evaluation Forum
3	2022	1.6	How "Loco" Is the LOCO Corpus? Annotating the Language of Conspiracy Theories (link)	Ludovic Mompelat et al.	Law
4	2021	8.7	LOCO: The 88-million-word language of conspiracy corpus (link)	A. Miani, T. Hills, and Adrian Bangerter	Behavior Research Methods
5	2024	5.4	Classifying Conspiratorial Narratives At Scale: False Alarms and Erroneous Connections (link)	Ahmad Diab, Rr. Nefriana, and Yu-Ru Lin	ArXiv
6	2024		What to Annotate: Retrieving Lexical Markers of Conspiracy Discourse from an Italian-English Corpus of Telegram Data (link)	C. Marini and Elisabetta Jezek	Proceedings of the 20th Joint ACL - ISO Workshop on Interoperable Semantic Annotation @ LREC-COLING 2024
7	2022	2.3	Hoaxes and Hidden agendas: A Twitter Conspiracy Theory Dataset: Data Paper (link)	S. C. Phillips, L. Ng, and K. Carley	Companion Proceedings of the Web Conference 2022
8	2023	12	COCO: an annotated Twitter dataset of COVID-19 conspiracy theories (link)	Johannes Langguth et al.	Journal of Computational Social Science
9	2024	0.9	Are You Serious? Handling Disagreement When Annotating Conspiracy Theory Texts (link)	Ashley Hemm et al.	Proceedings of the 18th Linguistic Annotation Workshop (LAW-XVIII)
10	2023	0.9	Bigfoot in Big Tech: Detecting Out of Domain Conspiracy Theories (link)	M. Fort et al.	Recent Advances in Natural Language Processing

	Year	Cit/yr	Title	Authors	Journal
11	2022	2.2	Which side are you on? Insider-Outsider classification in conspiracy-theoretic social media (link)	Pavan Holur, Tianyi Wang, Shadi Shahsavari, Timothy R. Tangherlini, and V. Roychowdhury	Annual Meeting of the Association for Computational Linguistics
12	2019	5.2	ŁThe Government Spies Using Our Webcams:ž The Language of Conspiracy Theories in Online Discussions (link)	Mattia Samory and Usa Virginia Tech	
13	2018	5.4	'The Government Spies Using Our Webcams' (link)	Mattia Samory and Tanushree Mitra	Proceedings of the ACM on Human-Computer Interaction
14	2020	18	An automated pipeline for the discovery of conspiracy and conspiracy theory narrative frameworks: Bridgegate, Pizzagate and storytelling on the web (link)	Timothy R. Tangherlini, Shadi Shahsavari, Behnam Shahbazi, Ehsan Ebrahimzadeh, and V. Roychowdhury	PLoS ONE
15	2020	44	Conspiracy in the time of corona: automatic detection of emerging COVID-19 conspiracy theories in social media and the news (link)	Shadi Shahsavari, Pavan Holur, Tianyi Wang, Timothy R. Tangherlini, and V. Roychowdhury	Journal of Computational Social Science
16	2022	0.8	Toward Detecting Conspiracy Language in Misinformation Documents (link)	Alana Platt, Jonathan Brown, and Amanda Venske	Proceedings of the 2022 Computers and People Research Conference
17	2023	4.5	Identifying Conspiracy Theories News based on Event Relation Graph (link)	Yuanyuan Lei and Ruihong Huang	ArXiv
18	2021	8.7	WICO Text: A Labeled Dataset of Conspiracy Theory and 5G-Corona Misinformation Tweets (link)	Konstantin Pogorelov, Daniel Thilo Schroeder, Petra Filkuková, Stefan Brenner, and J. Langguth	Proceedings of the 2021 Workshop on Open Challenges in Online Social Networks
19	2024	1.9	ConspEmoLLM: Conspiracy Theory Detection Using an Emotion-Based Large Language Model (link)	Zhiwei Liu et al.	ArXiv

	Year	Cit/yr	Title	Authors	Journal
20	2020	7.4	FakeNews: Corona Virus and 5G Conspiracy Task at MediaEval 2020 (link)	Konstantin Pogorelov et al.	MediaEval Benchmarking Initiative for Multimedia Evaluation
21	2021	5.5	FakeNews: Corona Virus and Conspiracies Multimedia Analysis Task at MediaEval 2021 (link)	Konstantin Pogorelov, Daniel Thilo Schroeder, Stefan Brenner, and J. Langguth	MediaEval Benchmarking Initiative for Multimedia Evaluation
22	2022	5.2	Combining Tweets and Connections Graph for FakeNews Detection at MediaEval 2022 (link)	Konstantin Pogorelov, Daniel Thilo Schroeder, Stefan Brenner, Asep Maulana, and Johannes Langguth	MediaEval Benchmarking Initiative for Multimedia Evaluation
23	2025		ConspirED: A Dataset for Cognitive Traits of Conspiracy Theories and Large Language Model Safety (link)	Luke Bates, Max Glockner, Preslav Nakov, and Iryna Gurevych	ArXiv
24	2023	2.5	ACTI at EVALITA 2023: Overview of the Conspiracy Theory Identification Task (link)	Giuseppe Russo, Niklas Stoehr, and Manoel Horta Ribeiro	ArXiv
25	2025	1.1	ConspEmoLLM-v2: A robust and stable model to detect sentiment-transformed conspiracy theories (link)	Zhiwei Liu, Paul Thompson, Jiaqi Rong, and Sophia Ananiadou	ArXiv
26	2021	4.0	Investigating Memorization of Conspiracy Theories in Text Generation (link)	Sharon Levy, Michael Stephen Saxon, and W. Wang	Findings
27	2025	2.0	The Schwurbelarchiv: a German Language Telegram dataset for the Study of Conspiracy Theories (link)	Mathias Angermaier, Elisabeth Hoeldrich, Jana Lasser, and Joao Pinheiro Neto	ArXiv
28	2023	0.7	ACTI at EVALITA 2023: Automatic Conspiracy Theory Identification Task Overview (short paper) (link)	Giuseppe Russo, Niklas Stoehr, and Manoel Horta Ribeiro	International Workshop on Evaluation of Natural Language and Speech Tools for Italian

	Year	Cit/yr	Title	Authors	Journal
29	2021	0.4	The Truth is Out There: Investigating Conspiracy Theories in Text Generation (link)	Sharon Levy, Michael Stephen Saxon, and William Yang Wang	ArXiv
30	2024	2.0	Unraveling the Italian and English Telegram Conspiracy Spheres through Message Forwarding (link)	Lorenzo Alvisi, S. Tardelli, and Maurizio Tesconi	ArXiv
31	2023	4.0	YouNICon: YouTube's CommuNIty of Conspiracy Videos (link)	Shaoyi Liaw, Fan Huang, Fabrício Benevenuto, Haewoon Kwak, and Jisun An	ArXiv
32	2024	2.6	Exploring the Usability of Persuasion Techniques for Downstream Misinformation-related Classification Tasks (link)	Nikolaos Nikolaidis, Jakub Piskorski, and Nicolas Stefanovitch	International Conference on Language Resources and Evaluation
33	2022	1.5	Using Persuasive Writing Strategies to Explain and Detect Health Misinformation (link)	Danial Kamali et al.	International Conference on Language Resources and Evaluation
34	2019	57	Fine-Grained Analysis of Propaganda in News Article (link)	Giovanni Da San Martino, Seunghak Yu, Alberto Barrón-Cedeño, R. Petrov, and Preslav Nakov	Conference on Empirical Methods in Natural Language Processing
35	2020	48	SemEval-2020 Task 11: Detection of Propaganda Techniques in News Articles (link)	Giovanni Da San Martino, A. Barr'on-Cedeno, Henning Wachsmuth, R. Petrov, and Preslav Nakov	International Workshop on Semantic Evaluation
36	2020	29	Fighting the COVID-19 Infodemic: Modeling the Perspective of Journalists, Fact-Checkers, Social Media Platforms, Policy Makers, and the Society (link)	Firoj Alam et al.	Conference on Empirical Methods in Natural Language Processing
37	2020	19	Fighting the COVID-19 Infodemic in Social Media: A Holistic Perspective and a Call to Arms (link)	Firoj Alam et al.	International Conference on Web and Social Media

	Year	Cit/yr	Title	Authors	Journal
38	2022	8.0	Empowering the Fact-checkers! Automatic Identification of Claim Spans on Twitter (link)	Megha Sundriyal, Atharva Kulkarni, Vaibhav Pulastya, Md. Shad Akhtar, and Tanmoy Chakraborty	ArXiv
39	2019	153	Understanding Conspiracy Theories (link)	Karen M. Douglas et al.	Political Psychology
40	2022	54	What Are Conspiracy Theories? A Definitional Approach to Their Correlates, Consequences, and Communication. (link)	Karen M. Douglas and Robbie M. Sutton	Annual review of psychology
41	2023	33	Conspiracy theories in online environments: An interdisciplinary literature review and agenda for future research (link)	Daniela Mahl, Mike S. Schäfer, and Jing Zeng	New Media & Society
42	2019	2.7	Conspiratorial Narratives in Violent Political Actors' Language (link)	Stephane J. Baele	Journal of Language and Social Psychology
43	2021	12	The language of conspiracy: A psychological analysis of speech used by conspiracy theorists and their followers on Twitter (link)	Amos Fong, J. Roozenbeek, Danielle Goldwert, Steven Rathje, and S. van der Linden	Group Processes & Intergroup Relations
44	2021	9.2	Detection of conspiracy propagators using psycho-linguistic characteristics (link)	Anastasia Giachanou, Bilal Ghanem, and Paolo Rosso	Journal of Information Science
45	2019	15	Pathways to conspiracy: The social and linguistic precursors of involvement in Reddit's conspiracy theory forum (link)	C. Klein, Peter Clutton, and A. Dunn	PLoS ONE
46	2022	4.5	Interconnectedness and (in)coherence as a signature of conspiracy worldviews (link)	A. Miani, T. Hills, and Adrian Bangerter	Science Advances
47	2013	9.5	"What about building 7?" A social psychological study of online discussion of 9/11 conspiracy theories (link)	M. Wood and Karen M. Douglas	Frontiers in Psychology

	Year	Cit/yr	Title	Authors	Journal
48	2018	16	Propagating and Debunking Conspiracy Theories on Twitter During the 2015–2016 Zika Virus Outbreak (link)	M. Wood	Cyberpsychology, Behavior and Social Networking
49	2021	5.7	Human Rationales as Attribution Priors for Explainable Stance Detection (link)	Sahil Jayaram and Emily Allaway	Conference on Empirical Methods in Natural Language Processing
50	2016	17	Rationale-Augmented Convolutional Neural Networks for Text Classification (link)	Ye Zhang, I. Marshall, and Byron C. Wallace	Proceedings of the Conference on Empirical Methods in Natural Language Processing. Conference on Empirical Methods in Natural Language Processing
51	2024	4.8	Human-annotated rationales and explainable text classification: a survey (link)	E. Herrewijnen, Dong Nguyen, Floris Bex, and K. V. Deemter	Frontiers in Artificial Intelligence

Paper Details

1. · 100% match · 2024 · 3.4 cit/yr

What distinguishes conspiracy from critical narratives? A computational analysis of oppositional discourse (link)

Damir Korenčić et al.

Expert Systems · Jul 9, 2024 · 6 citations

The current prevalence of conspiracy theories on the internet is a significant issue, tackled by many computational approaches. However, these approaches fail to recognize the relevance of distinguishing between texts which contain a conspiracy theory and texts which are simply critical and oppose mainstream narratives. Furthermore, little attention is usually paid to the role of inter-group conflict in oppositional narratives. We contribute by proposing a novel topic-agnostic annotation scheme that differentiates between conspiracies and critical texts, and that defines span-level categories of inter-group conflict. We also contribute with the multilingual XAI-DisInfodemics corpus (English and Spanish), which contains a high-quality annotation of Telegram messages related to COVID-19 (5000 messages per language). We also demonstrate the feasibility of an NLP-based automatization by performing a range of experiments that yield strong baseline solutions. Finally, we perform an analysis which demonstrates that the promotion of intergroup conflict and the presence of violence and anger are key aspects to distinguish between the two types of oppositional narratives, that is, conspiracy versus critical.

2. · 100% match · 2024 · 8.0 cit/yr
Overview of the Oppositional Thinking Analysis PAN Task at CLEF 2024 (link)
Damir Korenčić et al.
Conference and Labs of the Evaluation Forum · 18 citations

This paper describes the Oppositional Thinking Analysis task at CLEF 2024. The task focuses on analyzing conspiracy theories and critical thinking narratives, and is comprised of two subtasks. Subtask 1 is a binary classification task aimed at distinguishing between critical and conspiracy texts. Subtask 2 is a token classification task aimed at detecting text spans corresponding to the key elements of oppositional (critical and conspiracy) narratives. The subtasks are based on a dataset of English and Spanish COVID19-related texts obtained from oppositional Telegram channels, and labeled using a topic-agnostic annotation scheme [1]. A total of 82 teams participated in the challenge, and 17 teams published working notes papers with system descriptions. The participants employed a range of NLP methods and pushed the state-of-art performance on both subtasks beyond the performance of the strong baseline systems [1] that were provided.

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3. · 100% match · 2022 · 1.6 cit/yr
How "Loco" Is the LOCO Corpus? Annotating the Language of Conspiracy Theories (link)
Ludovic Mompelat et al.
Law · 7 citations

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4. · 100% match · 2021 · 8.7 cit/yr
LOCO: The 88-million-word language of conspiracy corpus (link)
A. Miani, T. Hills, and Adrian Bangerter
Behavior Research Methods · Oct 25, 2021 · 39 citations

The spread of online conspiracy theories represents a serious threat to society. To understand the content of conspiracies, here we present the language of conspiracy (LOCO) corpus. LOCO is an 88-million-token corpus composed of topic-matched conspiracy (N = 23,937) and mainstream (N = 72,806) documents harvested from 150 websites. Mimicking internet user behavior, documents were identified using Google by crossing a set of seed phrases with a set of websites. LOCO is hierarchically structured, meaning that each document is cross-nested within websites (N = 150) and topics (N = 600, on three different resolutions). A rich set of linguistic features (N = 287) and metadata includes upload date, measures of social media engagement, measures of website popularity, size, and traffic, as well as political bias and factual reporting annotations. We explored LOCO's features from different perspectives showing that documents track important societal events through time (e.g., Princess Diana's death, Sandy Hook school shooting, coronavirus outbreaks), while patterns of lexical features (e.g., deception, power, dominance) overlap with those extracted from online social media communities dedicated to conspiracy theories. By computing within-subcorpus cosine similarity, we derived a subset of the most representative conspiracy documents (N = 4,227), which, compared to other conspiracy documents, display prototypical and exaggerated conspiratorial language and are more frequently shared on Facebook. We also show that conspiracy website users navigate to websites via more direct means

than mainstream users, suggesting confirmation bias. LOCO and related datasets are freely available at https://osf.io/snpcg/.

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5. · 100% match · 2024 · 5.4 cit/yr

Classifying Conspiratorial Narratives At Scale: False Alarms and Erroneous Connections (link)

Ahmad Diab, Rr. Nefriana, and Yu-Ru Lin

ArXiv · Mar 29, 2024 · 11 citations

Online discussions frequently involve conspiracy theories, which can contribute to the proliferation of belief in them. However, not all discussions surrounding conspiracy theories promote them, as some are intended to debunk them. Existing research has relied on simple proxies or focused on a constrained set of signals to identify conspiracy theories, which limits our understanding of conspiratorial discussions across different topics and online communities. This work establishes a general scheme for classifying discussions related to conspiracy theories based on authors' perspectives on the conspiracy belief, which can be expressed explicitly through narrative elements, such as the agent, action, or objective, or implicitly through references to known theories, such as chemtrails or the New World Order. We leverage human-labeled ground truth to train a BERT-based model for classifying online CTs, which we then compared to the Generative Pre-trained Transformer machine (GPT) for detecting online conspiratorial content. Despite GPT's known strengths in its expressiveness and contextual understanding, our study revealed significant flaws in its logical reasoning, while also demonstrating comparable strengths from our classifiers. We present the first large-scale classification study using posts from the most active conspiracy-related Reddit forums and find that only one-third of the posts are classified as positive. This research sheds light on the potential applications of large language models in tasks demanding nuanced contextual comprehension.

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6. · 100% match · 2024

What to Annotate: Retrieving Lexical Markers of Conspiracy Discourse from an Italian-English Corpus of Telegram Data (link)

C. Marini and Elisabetta Jezek

Proceedings of the 20th Joint ACL - ISO Workshop on Interoperable Semantic Annotation @ LREC-COLING 2024 · May 1, 2024 · 0 citations

In this age of social media, Conspiracy Theories (CTs) have become an issue that can no longer be ignored. After providing an overview of CT literature and corpus studies, we describe the creation of a 40,000-token English-Italian bilingual corpus of conspiracy-oriented Telegram comments – the Complotto corpus – and the linguistic analysis we performed using the Sketch Engine online platform (Kilgarriff et al., 2010) on our annotated data to identify statistically relevant linguistic markers of CT discourse. Thanks to the platform's keywords and key terms extraction functions, we were able to assess the statistical significance of the following lexical and semantic phenomena, both cross-linguistically and cross-CT, namely: (1) evidentiality and epistemic modality markers; (2) debunking vocabulary referring to another version of the truth lying behind the official one; (3) the conceptual metaphor INSTITUTIONS ARE ABUSERS. All these features qualify as markers of CT discourse and

have the potential to be effectively used for future semantic annotation tasks to develop automatic systems for CT identification.

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7. · 100% match · 2022 · 2.3 cit/yr

Hoaxes and Hidden agendas: A Twitter Conspiracy Theory Dataset: Data Paper (link)

S. C. Phillips, L. Ng, and K. Carley

Companion Proceedings of the Web Conference 2022 · Apr 25, 2022 · 9 citations

Hoaxes and hidden agendas make for compelling conspiracy theories. While many of these theories are ultimately innocuous, others have the potential to do real harm, instigating real-world support or disapproval of the theories. This is further fueled by social media which provides a platform for conspiracy theories to spread at unprecedented rates. Thus, there is a need for the development of automated models to detect conspiracy theories from the social media space in order to quickly and effectively identify the topics of the season and the prevailing stance. To support this development, we create ground truth data through human annotation. In this work, we collect and manually annotate a dataset from Twitter, comprising of four conspiracy theories. Each Tweet is annotated with one of the four topics {climate change, COVID-19 origin, COVID-19 vaccine, Epstein-Maxwell trial}, and its stance towards the conspiracy theory {support, neutral, against}. We perform experiments on this multi-topic dataset to demonstrate its usage in conspiracy-detection, stance-detection and topic-detection.

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8. · 100% match · 2023 · 12 cit/yr

COCO: an annotated Twitter dataset of COVID-19 conspiracy theories (link)

Johannes Langguth et al.

Journal of Computational Social Science · Apr 4, 2023 · 37 citations

The COVID-19 pandemic has been accompanied by a surge of misinformation on social media which covered a wide range of different topics and contained many competing narratives, including conspiracy theories. To study such conspiracy theories, we created a dataset of 3495 tweets with manual labeling of the stance of each tweet w.r.t. 12 different conspiracy topics. The dataset thus contains almost 42,000 labels, each of which determined by majority among three expert annotators. The dataset was selected from COVID-19 related Twitter data spanning from January 2020 to June 2021 using a list of 54 keywords. The dataset can be used to train machine learning based classifiers for both stance and topic detection, either individually or simultaneously. BERT was used successfully for the combined task. The dataset can also be used to further study the prevalence of different conspiracy narratives. To this end we qualitatively analyze the tweets, discussing the structure of conspiracy narratives that are frequently found in the dataset. Furthermore, we illustrate the interconnection between the conspiracy categories as well as the keywords.

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9. · 100% match · 2024 · 0.9 cit/yr

Are You Serious? Handling Disagreement When Annotating Conspiracy Theory Texts (link)

Ashley Hemm et al.

Proceedings of the 18th Linguistic Annotation Workshop (LAW-XVIII) · 2 citations

We often assume that annotation tasks, such as annotating for the presence of conspiracy theories, can be annotated with hard labels, without definitions or guidelines. Our annotation experiments, comparing students and experts, show that there is little agreement on basic annotations even among experts. For this reason, we conclude that we need to accept disagreement as an integral part of such annotations.

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10. · 100% match · 2023 · 0.9 cit/yr

Bigfoot in Big Tech: Detecting Out of Domain Conspiracy Theories (link)

M. Fort et al.

Recent Advances in Natural Language Processing · 3 citations

We investigate approaches to classifying texts into either conspiracy theory or mainstream using the Language Of Conspiracy (LOCO) corpus. Since conspiracy theories are not monolithic constructs, we need to identify approaches that robustly work in an out-of- domain setting (i.e., across conspiracy topics). We investigate whether optimal in-domain set- tings can be transferred to out-of-domain settings, and we investigate different methods for bleaching to steer classifiers away from words typical for an individual conspiracy theory. We find that BART works better than an SVM, that we can successfully classify out-of-domain, but there are no clear trends in how to choose the best source training domains. Addition- ally, bleaching only topic words works better than bleaching all content words or completely delexicalizing texts.

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11. · 100% match · 2022 · 2.2 cit/yr

Which side are you on? Insider-Outsider classification in conspiracy-theoretic social media (link)

Pavan Holur, Tianyi Wang, Shadi Shahsavari, Timothy R. Tangherlini, and V. Roychowdhury

Annual Meeting of the Association for Computational Linguistics · Mar 8, 2022 · 9 citations

Social media is a breeding ground for threat narratives and related conspiracy theories. In these, an outside group threatens the integrity of an inside group, leading to the emergence of sharply defined group identities: Insiders – agents with whom the authors identify and Outsiders – agents who threaten the insiders. Inferring the members of these groups constitutes a challenging new NLP task: (i) Information is distributed over many poorly-constructed posts; (ii) Threats and threat agents are highly contextual, with the same post potentially having multiple agents assigned to membership in either group; (iii) An agent's identity is often implicit and transitive; and (iv) Phrases used to imply Outsider status often do not follow common negative sentiment patterns. To address these challenges, we define a novel Insider-Outsider classification task. Because we are not aware of any appropriate existing datasets or attendant models, we introduce a labeled dataset (CT5K) and design a model (NP2IO) to address this task. NP2IO leverages pretrained language modeling to classify Insiders and Outsiders. NP2IO is shown to be robust, generalizing to noun phrases not seen during training, and exceeding the performance of non-trivial baseline models by 20%.

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12. · 100% match · 2019 · 5.2 cit/yr

łThe Government Spies Using Our Webcams:ž The Language of Conspiracy Theories in Online Discussions (link)

Mattia Samory and Usa Virginia Tech

38 citations

Conspiracy theories are omnipresent in online discussionsÐwhether to explain a late-breaking event that still lacks official report or to give voice to political dissent. Conspiracy theories evolve, multiply, and interconnect, further complicating efforts to limit their propagation. It is therefore crucial to develop scalable methods to examine the nature of conspiratorial discussions in online communities. What do users talk about when they discuss conspiracy theories online? What are the recurring elements in their discussions? What do these elements tell us about the way users think? This work answers these questions by analyzing over ten years of discussions in r/conspiracyÐan online community on Reddit dedicated to conspiratorial discussions. We focus on the key elements of a conspiracy theory: the conspiratorial agents , the actions they perform, and their targets . By computationally detecting agentśactionśtarget triplets in conspiratorial statements, and grouping them into semantically coherent clusters, we develop a notion of narrative-motif to detect recurring patterns of triplets. For example, a narrative-motif such as łgovernmental agencyścontrolsścommunicationsž appears in diverse conspiratorial statements alleging that governmental agencies control information to nefarious ends. Thus, narrative-motifs expose commonalities between multiple conspiracy theories even when they refer to different events or circumstances. In the process, these representations help us understand how users talk about conspiracy theories and offer us a means to interpret what they talk about. Our approach enables a population-scale study of conspiracy theories in alternative news and social media with implications for understanding their adoption and combating their spread.

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13. · 100% match · 2018 · 5.4 cit/yr

'The Government Spies Using Our Webcams' (link)

Mattia Samory and Tanushree Mitra

Proceedings of the ACM on Human-Computer Interaction · Nov 1, 2018 · 40 citations

Conspiracy theories are omnipresent in online discussions—whether to explain a late-breaking event that still lacks official report or to give voice to political dissent. Conspiracy theories evolve, multiply, and interconnect, further complicating efforts to understand them and to limit their propagation. It is therefore crucial to develop scalable methods to examine the nature of conspiratorial discussions in online communities. What do users talk about when they discuss conspiracy theories online? What are the recurring elements in their discussions? What do these elements tell us about the way users think? This work answers these questions by analyzing over ten years of discussions in r/conspiracy—an online community on Reddit dedicated to conspiratorial discussions. We focus on the key elements of a conspiracy theory: the conspiratorial agents, the actions they perform, and their targets. By computationally detecting agent?action?target triplets in conspiratorial statements, and grouping them into semantically coherent clusters, we develop a notion of narrative-motif to detect recurring patterns of triplets. For example, a narrative-motif such as "governmental agency-controls-communications" represents the various ways in which multiple conspiratorial statements denote how

governmental agencies control information. Thus, narrative-motifs expose commonalities between multiple conspiracy theories even when they refer to different events or circumstances. In the process, these representations help us understand how users talk about conspiracy theories and offer us a means to interpret what they talk about. Our approach enables a population-scale study of conspiracy theories in alternative news and social media with implications for understanding their adoption and combating their spread.

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14. · 100% match · 2020 · 18 cit/yr

An automated pipeline for the discovery of conspiracy and conspiracy theory narrative frameworks: Bridgegate, Pizzagate and storytelling on the web (link)

Timothy R. Tangherlini, Shadi Shahsavari, Behnam Shahbazi, Ehsan Ebrahimzadeh, and V. Roychowdhury

PLoS ONE · Jun 16, 2020 · 104 citations

Although a great deal of attention has been paid to how conspiracy theories circulate on social media, and the deleterious effect that they, and their factual counterpart conspiracies, have on political institutions, there has been little computational work done on describing their narrative structures. Predicating our work on narrative theory, we present an automated pipeline for the discovery and description of the generative narrative frameworks of conspiracy theories that circulate on social media, and actual conspiracies reported in the news media. We base this work on two separate comprehensive repositories of blog posts and news articles describing the well-known conspiracy theory Pizzagate from 2016, and the New Jersey political conspiracy Bridgegate from 2013. Inspired by the qualitative narrative theory of Greimas, we formulate a graphical generative machine learning model where nodes represent actors/actants, and multi-edges and self-loops among nodes capture context-specific relationships. Posts and news items are viewed as samples of subgraphs of the hidden narrative framework network. The problem of reconstructing the underlying narrative structure is then posed as a latent model estimation problem. To derive the narrative frameworks in our target corpora, we automatically extract and aggregate the actants (people, places, objects) and their relationships from the posts and articles. We capture context specific actants and interactant relationships by developing a system of supernodes and subnodes. We use these to construct an actant-relationship network, which constitutes the underlying generative narrative framework for each of the corpora. We show how the Pizzagate framework relies on the conspiracy theorists' interpretation of "hidden knowledge" to link otherwise unlinked domains of human interaction, and hypothesize that this multi-domain focus is an important feature of conspiracy theories. We contrast this to the single domain focus of an actual conspiracy. While Pizzagate relies on the alignment of multiple domains, Bridgegate remains firmly rooted in the single domain of New Jersey politics. We hypothesize that the narrative framework of a conspiracy theory might stabilize quickly in contrast to the narrative framework of an actual conspiracy, which might develop more slowly as revelations come to light. By highlighting the structural differences between the two narrative frameworks, our approach could be used by private and public analysts to help distinguish between conspiracy theories and conspiracies.

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15. · 100% match · 2020 · 44 cit/yr

Conspiracy in the time of corona: automatic detection of emerging COVID-19 conspiracy theories in social media and the news (link)

Shadi Shahsavari, Pavan Holur, Tianyi Wang, Timothy R. Tangherlini, and V. Roychowdhury

Journal of Computational Social Science · Aug 4, 2020 · 250 citations

Rumors and conspiracy theories thrive in environments of low confidence and low trust. Consequently, it is not surprising that ones related to the COVID-19 pandemic are proliferating given the lack of scientific consensus on the virus's spread and containment, or on the long-term social and economic ramifications of the pandemic. Among the stories currently circulating in US-focused social media forums are ones suggesting that the 5G telecommunication network activates the virus, that the pandemic is a hoax perpetrated by a global cabal, that the virus is a bio-weapon released deliberately by the Chinese, or that Bill Gates is using it as cover to launch a broad vaccination program to facilitate a global surveillance regime. While some may be quick to dismiss these stories as having little impact on real-world behavior, recent events including the destruction of cell phone towers, racially fueled attacks against Asian Americans, demonstrations espousing resistance to public health orders, and wide-scale defiance of scientifically sound public mandates such as those to wear masks and practice social distancing, countermand such conclusions. Inspired by narrative theory, we crawl social media sites and news reports and, through the application of automated machine-learning methods, discover the underlying narrative frameworks supporting the generation of rumors and conspiracy theories. We show how the various narrative frameworks fueling these stories rely on the alignment of otherwise disparate domains of knowledge, and consider how they attach to the broader reporting on the pandemic. These alignments and attachments, which can be monitored in near real time, may be useful for identifying areas in the news that are particularly vulnerable to reinterpretation by conspiracy theorists. Understanding the dynamics of storytelling on social media and the narrative frameworks that provide the generative basis for these stories may also be helpful for devising methods to disrupt their spread.

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16. · 100% match · 2022 · 0.8 cit/yr

Toward Detecting Conspiracy Language in Misinformation Documents (link)

Alana Platt, Jonathan Brown, and Amanda Venske

Proceedings of the 2022 Computers and People Research Conference · Jun 2, 2022 · 3 citations

Recent events have illustrated the danger of online conspiracy theories to cause harm in the real world. In this research in progress, we focus on machine learning techniques to differentiate conspiracy information from other forms of misinformation. Our results demonstrate that conspiracy language is differentiable and suggest there may exist features of conspiracy language independent of the conspiracy topic. We also present a direction for future work to better understand the unique features of conspiracy language and how they may be used to enhance machine learning techniques.

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17. · 100% match · 2023 · 4.5 cit/yr

Identifying Conspiracy Theories News based on Event Relation Graph (link)

Yuanyuan Lei and Ruihong Huang
ArXiv · Oct 28, 2023 · 11 citations

Conspiracy theories, as a type of misinformation, are narratives that explains an event or situation in an irrational or malicious manner. While most previous work examined conspiracy theory in social media short texts, limited attention was put on such misinformation in long news documents. In this paper, we aim to identify whether a news article contains conspiracy theories. We observe that a conspiracy story can be made up by mixing uncorrelated events together, or by presenting an unusual distribution of relations between events. Achieving a contextualized understanding of events in a story is essential for detecting conspiracy theories. Thus, we propose to incorporate an event relation graph for each article, in which events are nodes, and four common types of event relations, coreference, temporal, causal, and subevent relations, are considered as edges. Then, we integrate the event relation graph into conspiracy theory identification in two ways: an event-aware language model is developed to augment the basic language model with the knowledge of events and event relations via soft labels; further, a heterogeneous graph attention network is designed to derive a graph embedding based on hard labels. Experiments on a large benchmark dataset show that our approach based on event relation graph improves both precision and recall of conspiracy theory identification, and generalizes well for new unseen media sources.

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18. · 100% match · 2021 · 8.7 cit/yr

WICO Text: A Labeled Dataset of Conspiracy Theory and 5G-Corona Misinformation Tweets (link)

Konstantin Pogorelov, Daniel Thilo Schroeder, Petra Filkuková, Stefan Brenner, and J. Langguth
Proceedings of the 2021 Workshop on Open Challenges in Online Social Networks · Aug 30, 2021 · 40 citations

The COVID-19 pandemic has been accompanied by a flood of misinformation on social media, which has been labeled an “infodemic”. While a large part of such fake news is ultimately inconsequential, some of it has the potential to real-world harm, but due to the massive amount of social media contents, it is impossible to find this misinformation manually. Thus, conventional fact-checking can typically only counteract misinformation narratives after they have gained significant traction. Only automated systems can provide warnings in advance. However, the automatic detection of misinformation narratives is very challenging since the texts that spread misinformation may be short messages on Twitter. They may also transmit misinformation by implication rather than by stating counterfactual information outright, and satirical messages complicate the issue further. Thus, there is a need for highly sophisticated detection systems. In order to support their development, we created substantial ground truth data by human annotation. In this paper, we present a dataset that deals with a specific piece of misinformation: the idea that the COVID-19 pandemic is causally connected to the 5G wireless network. We selected more than 10,000 tweets that deal with COVID-19 and 5G and labeled them manually, distinguishing between tweets that propagate the specific 5G misinformation, those that spread other conspiracy theories, and tweets that do neither. We provide the human-annotated dataset along with an additional large-scale automatically (by using the human-annotated dataset as the training set) labelled dataset consist of more than 100,000 tweets.

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19. · 100% match · 2024 · 1.9 cit/yr

ConspEmoLLM: Conspiracy Theory Detection Using an Emotion-Based Large Language Model (link)

Zhiwei Liu et al.

ArXiv · Mar 11, 2024 · 4 citations

The internet has brought both benefits and harms to society. A prime example of the latter is misinformation, including conspiracy theories, which flood the web. Recent advances in natural language processing, particularly the emergence of large language models (LLMs), have improved the prospects of accurate misinformation detection. However, most LLM-based approaches to conspiracy theory detection focus only on binary classification and fail to account for the important relationship between misinformation and affective features (i.e., sentiment and emotions). Driven by a comprehensive analysis of conspiracy text that reveals its distinctive affective features, we propose ConspEmoLLM, the first open-source LLM that integrates affective information and is able to perform diverse tasks relating to conspiracy theories. These tasks include not only conspiracy theory detection, but also classification of theory type and detection of related discussion (e.g., opinions towards theories). ConspEmoLLM is fine-tuned based on an emotion-oriented LLM using our novel ConDID dataset, which includes five tasks to support LLM instruction tuning and evaluation. We demonstrate that when applied to these tasks, ConspEmoLLM largely outperforms several open-source general domain LLMs and ChatGPT, as well as an LLM that has been fine-tuned using ConDID, but which does not use affective features. This project will be released on https://github.com/lzw108/ConspEmoLLM/.

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20. · 100% match · 2020 · 7.4 cit/yr

FakeNews: Corona Virus and 5G Conspiracy Task at MediaEval 2020 (link)

Konstantin Pogorelov et al.

MediaEval Benchmarking Initiative for Multimedia Evaluation · 46 citations

The FakeNews: Corona Virus and 5G Conspiracy task, running for the first time as part of MediaEval 2020, focuses on the classification of tweet texts and retweet cascades for the detection of fast-spreading misinformation, and therefore provides a low-threshold introduction to natural language processing and graph analysis. This paper describes the task, including use case and motivation, challenges, the dataset with ground truth, the required participant runs, and the evaluation metrics.

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21. · 100% match · 2021 · 5.5 cit/yr

FakeNews: Corona Virus and Conspiracies Multimedia Analysis Task at MediaEval 2021 (link)

Konstantin Pogorelov, Daniel Thilo Schroeder, Stefan Brenner, and J. Langguth

MediaEval Benchmarking Initiative for Multimedia Evaluation · 29 citations

The FakeNews: Corona Virus and Conspiracies Multimedia Analysis task, running for the second time as part of MediaEval 2021, focuses on the classification of tweet texts aiming detection of fast-spreading misinformation. Task of this year extends the number of target conspiracy theories and introduces new challenges in terms of analysis complexity of the imbalanced dataset. This paper describes the task,

including use case and motivation, challenges, the dataset with ground truth, the required participant runs, and the evaluation metrics.

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22. · 100% match · 2022 · 5.2 cit/yr

Combining Tweets and Connections Graph for FakeNews Detection at MediaEval 2022 (link)

Konstantin Pogorelov, Daniel Thilo Schroeder, Stefan Brenner, Asep Maulana, and Johannes Langguth

MediaEval Benchmarking Initiative for Multimedia Evaluation · 22 citations

The FakeNews Detection task at MediaEval 2022, running for the third time as part of the challenge, focuses on the detection of misinformation tweets and their spreaders. Like in the 2021 task, conspiracy theories related to COVID-19 in nine different categories have to be detected, along with the authors stance towards them. For the 2022 challenge, the size of the dataset has approximately doubled. Furthermore, we also provide a large interaction graph along with vertex features derived from the same Twitter dataset in which misinformation spreaders should be classified. As a final subtask, participants are asked to combine text and graph information to refine their classifications. This paper describes the tasks, including use case and motivation, challenges, the dataset with ground truth, the required participant runs, and the evaluation metrics.

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23. · 100% match · 2025

ConspirED: A Dataset for Cognitive Traits of Conspiracy Theories and Large Language Model Safety (link)

Luke Bates, Max Glockner, Preslav Nakov, and Iryna Gurevych

ArXiv · Aug 28, 2025 · 0 citations

Conspiracy theories erode public trust in science and institutions while resisting debunking by evolving and absorbing counter-evidence. As AI-generated misinformation becomes increasingly sophisticated, understanding rhetorical patterns in conspiratorial content is important for developing interventions such as targeted prebunking and assessing AI vulnerabilities. We introduce ConspirED (CONSPIR Evaluation Dataset), which captures the cognitive traits of conspiratorial ideation in multi-sentence excerpts (80–120 words) from online conspiracy articles, annotated using the CONSPIR cognitive framework (Lewandowsky and Cook, 2020). ConspirED is the first dataset of conspiratorial content annotated for general cognitive traits. Using ConspirED, we (i) develop computational models that identify conspiratorial traits and determine dominant traits in text excerpts, and (ii) evaluate large language/reasoning model (LLM/LRM) robustness to conspiratorial inputs. We find that both are misaligned by conspiratorial content, producing output that mirrors input reasoning patterns, even when successfully deflecting comparable fact-checked misinformation.

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24. · 100% match · 2023 · 2.5 cit/yr

ACTI at EVALITA 2023: Overview of the Conspiracy Theory Identification Task (link)

Giuseppe Russo, Niklas Stoehr, and Manoel Horta Ribeiro

ArXiv · 8 citations

Automatic Conspiracy Theory Identification (ACTI) is a new shared task proposed for the first time at the EVALITA 2023 evaluation campaign. ACTI is based on a new, manually labeled dataset of comments scraped from conspiratorial Telegram channels and consists of two subtasks: (1) identifying conspiratorial content (conspiratorial content classification); and (2) classifying content into specific con-spiracy theories (conspiratorial category classification). A total of 15 teams participated in the task with 81 submissions. In this task sum-mary, we discuss the data and task, and outline the best-performing approaches that are largely based on large language models. We conclude with a brief discussion of the application of large language models to counter the spread of misinformation on online platforms.

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25. · 100% match · 2025 · 1.1 cit/yr

ConspEmoLLM-v2: A robust and stable model to detect sentiment-transformed conspiracy theories (link)

Zhiwei Liu, Paul Thompson, Jiaqi Rong, and Sophia Ananiadou

ArXiv · May 20, 2025 · 1 citations

Despite the many benefits of large language models (LLMs), they can also cause harm, e.g., through automatic generation of misinformation, including conspiracy theories. Moreover, LLMs can also''disguise''conspiracy theories by altering characteristic textual features, e.g., by transforming their typically strong negative emotions into a more positive tone. Although several studies have proposed automated conspiracy theory detection methods, they are usually trained using human-authored text, whose features can vary from LLM-generated text. Furthermore, several conspiracy detection models, including the previously proposed ConspEmoLLM, rely heavily on the typical emotional features of human-authored conspiracy content. As such, intentionally disguised content may evade detection. To combat such issues, we firstly developed an augmented version of the ConDID conspiracy detection dataset, ConDID-v2, which supplements human-authored conspiracy tweets with versions rewritten by an LLM to reduce the negativity of their original sentiment. The quality of the rewritten tweets was verified by combining human and LLM-based assessment. We subsequently used ConDID-v2 to train ConspEmoLLM-v2, an enhanced version of ConspEmoLLM. Experimental results demonstrate that ConspEmoLLM-v2 retains or exceeds the performance of ConspEmoLLM on the original human-authored content in ConDID, and considerably outperforms both ConspEmoLLM and several other baselines when applied to sentiment-transformed tweets in ConDID-v2. The project will be available at https://github.com/lzw108/ConspEmoLLM.

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26. · 100% match · 2021 · 4.0 cit/yr

Investigating Memorization of Conspiracy Theories in Text Generation (link)

Sharon Levy, Michael Stephen Saxon, and W. Wang

Findings · Jan 2, 2021 · 21 citations

The adoption of natural language generation (NLG) models can leave individuals vulnerable to the generation of harmful information memorized by the models, such as conspiracy theories. While previous studies examine conspiracy theories in the context of social media, they have not evaluated their

presence in the new space of generative language models. In this work, we investigate the capability of language models to generate conspiracy theory text. Specifically, we aim to answer: can we test pretrained generative language models for the memorization and elicitation of conspiracy theories without access to the model's training data? We highlight the difficulties of this task and discuss it in the context of memorization, generalization, and hallucination. Utilizing a new dataset consisting of conspiracy theory topics and machine-generated conspiracy theories helps us discover that many conspiracy theories are deeply rooted in the pretrained language models. Our experiments demonstrate a relationship between model parameters such as size and temperature and their propensity to generate conspiracy theory text. These results indicate the need for a more thorough review of NLG applications before release and an in-depth discussion of the drawbacks of memorization in generative language models.

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27. · 100% match · 2025 · 2.0 cit/yr

The Schwurbelarchiv: a German Language Telegram dataset for the Study of Conspiracy Theories (link)

Mathias Angermaier, Elisabeth Hoeldrich, Jana Lasser, and Joao Pinheiro Neto

ArXiv · Apr 8, 2025 · 2 citations

Sociality borne by language, as is the predominant digital trace on text-based social media platforms, harbours the raw material for exploring multiple social phenomena. Distinctively, the messaging service Telegram provides functionalities that allow for socially interactive as well as one-to-many communication. Our Telegram dataset contains over 6,000 groups and channels, 40 million text messages, and over 3 million transcribed audio files, originating from a data-hoarding initiative named the ``Schwurbelarchiv'' (from German schwurbeln: speaking nonsense). This dataset publication details the structure, scope, and methodological specifics of the Schwurbelarchiv, emphasising its relevance for further research on the German-language conspiracy theory discourse. We validate its predominantly German origin by linguistic and temporal markers and situate it within the context of similar datasets. We describe process and extent of the transcription of multimedia files. Thanks to this effort the dataset uniquely supports multimodal analysis of online social dynamics and content dissemination. Researchers can employ this resource to explore societal dynamics in misinformation, political extremism, opinion adaptation, and social network structures on Telegram. The Schwurbelarchiv thus offers unprecedented opportunities for investigations into digital communication and its societal implications.

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28. · 100% match · 2023 · 0.7 cit/yr

ACTI at EVALITA 2023: Automatic Conspiracy Theory Identification Task Overview (short paper) (link)

Giuseppe Russo, Niklas Stoehr, and Manoel Horta Ribeiro

International Workshop on Evaluation of Natural Language and Speech Tools for Italian · Jul 12, 2023 · 2 citations

Conspiracy Theory Identication task is a new shared task proposed for the first time at the Evalita 2023. The ACTI challenge, based exclusively on comments published on conspiratorial channels of telegram, is divided into two subtasks: (i) Conspiratorial Content Classification: identifying conspiratorial content and (ii) Conspiratorial Category Classification about specific conspiracy theory classification. A total of fifteen teams participated in the task for a total of 81 submissions. We illustrate the best performing approaches were based on the utilization of large language models. We finally draw conclusions about the utilization of these models for counteracting the spreading of misinformation in online platforms.

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29. · 100% match · 2021 · 0.4 cit/yr

The Truth is Out There: Investigating Conspiracy Theories in Text Generation (link)

Sharon Levy, Michael Stephen Saxon, and William Yang Wang

ArXiv · 2 citations

With the growing adoption of text generation models in today's society, users are increasingly exposed to machine-generated text. This in turn can leave users vulnerable to the generation of harmful information such as con-spiracy theories. While the propagation of conspiracy theories through social media has been studied, previous work has not evaluated their diffusion through text generation. In this work, we investigate the propensity for language models to generate conspiracy theory text. Our study focuses on testing these models for the elicitation of conspiracy theories and comparing these generations to human-written theories from Reddit. We also introduce a new dataset consisting of conspiracy theory topics, machine-generated conspiracy theories, and human-written conspiracy theories. Our experiments show that many well-known conspiracy theory topics are deeply rooted in the pre-trained language models, and can become more prevalent through different model settings.

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30. · 100% match · 2024 · 2.0 cit/yr

Unraveling the Italian and English Telegram Conspiracy Spheres through Message Forwarding (link)

Lorenzo Alvisi, S. Tardelli, and Maurizio Tesconi

ArXiv · Apr 29, 2024 · 4 citations

Telegram has grown into a significant platform for news and information sharing, favored for its anonymity and minimal moderation. This openness, however, makes it vulnerable to misinformation and conspiracy theories. In this study, we explore the dynamics of conspiratorial narrative dissemination within Telegram, focusing on Italian and English landscapes. In particular, we leverage the mechanism of message forwarding within Telegram and collect two extensive datasets through snowball strategy. We adopt a network-based approach and build the Italian and English Telegram networks to reveal their respective communities. By employing topic modeling, we uncover distinct narratives and dynamics of misinformation spread. Results highlight differences between Italian and English conspiracy landscapes, with Italian discourse involving assorted conspiracy theories and alternative news sources intertwined with legitimate news sources, whereas English discourse is characterized by a more focused approach on specific narratives such as QAnon and political conspiracies. Finally,

we show that our methodology exhibits robustness across initial seed selections, suggesting broader applicability. This study contributes to understanding information and misinformation spread on Italian and English Telegram ecosystems through the mechanism of message forwarding

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31. · 100% match · 2023 · 4.0 cit/yr

YouNICon: YouTube's CommuNIty of Conspiracy Videos (link)

Shaoyi Liaw, Fan Huang, Fabrício Benevenuto, Haewoon Kwak, and Jisun An

ArXiv · Apr 11, 2023 · 12 citations

Conspiracy theories are widely propagated on social media. Among various social media services, YouTube is one of the most influential sources of news and entertainment. This paper seeks to develop a dataset, YOUNICON, to enable researchers to perform conspiracy theory detection as well as classification of videos with conspiracy theories into different topics. YOUNICON is a dataset with a large collection of videos from suspicious channels that were identified to contain conspiracy theories in a previous study. Overall, YOUNICON will enable researchers to study trends in conspiracy theories and understand how individuals can interact with the conspiracy theory producing community or channel. Our data is available at: https://doi.org/10.5281/zenodo.7466262.

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32. · 100% match · 2024 · 2.6 cit/yr

Exploring the Usability of Persuasion Techniques for Downstream Misinformation-related Classification Tasks (link)

Nikolaos Nikolaidis, Jakub Piskorski, and Nicolas Stefanovitch

International Conference on Language Resources and Evaluation · May 1, 2024 · 5 citations

We systematically explore the predictive power of features derived from Persuasion Techniques detected in texts, for solving different tasks of interest for media analysis; notably: detecting mis/disinformation, fake news, propaganda, partisan news and conspiracy theories. Firstly, we propose a set of meaningful features, aiming to capture the persuasiveness of a text. Secondly, we assess the discriminatory power of these features in different text classification tasks on 8 selected datasets from the literature using two metrics. We also evaluate the per-task discriminatory power of each Persuasion Technique and report on different insights. We find out that most of these features have a noticeable potential to distinguish conspiracy theories, hyperpartisan news and propaganda, while we observed mixed results in the context of fake news detection.

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33. · 100% match · 2022 · 1.5 cit/yr

Using Persuasive Writing Strategies to Explain and Detect Health Misinformation (link)

Danial Kamali et al.

International Conference on Language Resources and Evaluation · Nov 11, 2022 · 5 citations

Nowadays, the spread of misinformation is a prominent problem in society. Our research focuses on aiding the automatic identification of misinformation by analyzing the persuasive strategies employed in textual documents. We introduce a novel annotation scheme encompassing common persuasive

writing tactics to achieve our objective. Additionally, we provide a dataset on health misinformation, thoroughly annotated by experts utilizing our proposed scheme. Our contribution includes proposing a new task of annotating pieces of text with their persuasive writing strategy types. We evaluate fine-tuning and prompt-engineering techniques with pre-trained language models of the BERT family and the generative large language models of the GPT family using persuasive strategies as an additional source of information. We evaluate the effects of employing persuasive strategies as intermediate labels in the context of misinformation detection. Our results show that those strategies enhance accuracy and improve the explainability of misinformation detection models. The persuasive strategies can serve as valuable insights and explanations, enabling other models or even humans to make more informed decisions regarding the trustworthiness of the information.

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34. · 95% match · 2019 · 57 cit/yr

Fine-Grained Analysis of Propaganda in News Article (link)

Giovanni Da San Martino, Seunghak Yu, Alberto Barrón-Cedeño, R. Petrov, and Preslav Nakov

Conference on Empirical Methods in Natural Language Processing · Oct 1, 2019 · 373 citations

Propaganda aims at influencing people's mindset with the purpose of advancing a specific agenda. Previous work has addressed propaganda detection at document level, typically labelling all articles from a propagandistic news outlet as propaganda. Such noisy gold labels inevitably affect the quality of any learning system trained on them. A further issue with most existing systems is the lack of explainability. To overcome these limitations, we propose a novel task: performing fine-grained analysis of texts by detecting all fragments that contain propaganda techniques as well as their type. In particular, we create a corpus of news articles manually annotated at fragment level with eighteen propaganda techniques and propose a suitable evaluation measure. We further design a novel multi-granularity neural network, and we show that it outperforms several strong BERT-based baselines.

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35. · 92% match · 2020 · 48 cit/yr

SemEval-2020 Task 11: Detection of Propaganda Techniques in News Articles (link)

Giovanni Da San Martino, A. Barr'on-Cedeno, Henning Wachsmuth, R. Petrov, and Preslav Nakov

International Workshop on Semantic Evaluation · Sep 6, 2020 · 266 citations

We present the results and the main findings of SemEval-2020 Task 11 on Detection of Propaganda Techniques in News Articles. The task featured two subtasks. Subtask SI is about Span Identification: given a plain-text document, spot the specific text fragments containing propaganda. Subtask TC is about Technique Classification: given a specific text fragment, in the context of a full document, determine the propaganda technique it uses, choosing from an inventory of 14 possible propaganda techniques. The task attracted a large number of participants: 250 teams signed up to participate and 44 made a submission on the test set. In this paper, we present the task, analyze the results, and discuss the system submissions and the methods they used. For both subtasks, the best systems used pre-trained Transformers and ensembles.

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36. · 90% match · 2020 · 29 cit/yr

Fighting the COVID-19 Infodemic: Modeling the Perspective of Journalists, Fact-Checkers, Social Media Platforms, Policy Makers, and the Society (link)

Firoj Alam et al.

Conference on Empirical Methods in Natural Language Processing · Apr 30, 2020 · 170 citations

With the emergence of the COVID-19 pandemic, the political and the medical aspects of disinformation merged as the problem got elevated to a whole new level to become the first global infodemic. Fighting this infodemic is ranked second in the list of the most important focus areas of the World Health Organization, with dangers ranging from promoting fake cures, rumors, and conspiracy theories to spreading xenophobia and panic. Addressing the issue requires solving a number of challenging problems such as identifying messages containing claims, determining their check-worthiness and factuality, and their potential to do harm as well as the nature of that harm, to mention just a few. Thus, here we design, annotate, and release to the research community a new dataset for fine-grained disinformation analysis that (i)focuses on COVID-19, (ii) combines the perspectives and the interests of journalists, fact-checkers, social media platforms, policy makers, and society as a whole, and (iii) covers both English and Arabic. Finally, we show strong evaluation results using state-of-the-art Transformers, thus confirming the practical utility of the annotation schema and of the dataset.

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37. · 72% match · 2020 · 19 cit/yr

Fighting the COVID-19 Infodemic in Social Media: A Holistic Perspective and a Call to Arms (link)

Firoj Alam et al.

International Conference on Web and Social Media · Jul 15, 2020 · 110 citations

With the outbreak of the COVID-19 pandemic, people turned to social media to read and to share timely information including statistics, warnings, advice, and inspirational stories. Unfortunately, alongside all this useful information, there was also a new blending of medical and political misinformation and disinformation, which gave rise to the first global infodemic. While fighting this infodemic is typically thought of in terms of factuality, the problem is much broader as malicious content includes not only fake news, rumors, and conspiracy theories, but also promotion of fake cures, panic, racism, xenophobia, and mistrust in the authorities, among others. This is a complex problem that needs a holistic approach combining the perspectives of journalists, fact-checkers, policymakers, government entities, social media platforms, and society as a whole. With this in mind, we define an annotation schema and detailed annotation instructions that reflect these perspectives. We further deploy a multilingual annotation platform, and we issue a call to arms to the research community and beyond to join the fight by supporting our crowdsourcing annotation efforts. We perform initial annotations using the annotation schema, and our initial experiments demonstrated sizable improvements over the baselines.

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38. · 66% match · 2022 · 8.0 cit/yr

Empowering the Fact-checkers! Automatic Identification of Claim Spans on Twitter (link)

Megha Sundriyal, Atharva Kulkarni, Vaibhav Pulastya, Md. Shad Akhtar, and Tanmoy Chakraborty
ArXiv · Oct 10, 2022 · 28 citations

The widespread diffusion of medical and political claims in the wake of COVID-19 has led to a voluminous rise in misinformation and fake news. The current vogue is to employ manual fact-checkers to efficiently classify and verify such data to combat this avalanche of claim-ridden misinformation. However, the rate of information dissemination is such that it vastly outpaces the fact-checkers' strength. Therefore, to aid manual fact-checkers in eliminating the superfluous content, it becomes imperative to automatically identify and extract the snippets of claim-worthy (mis)information present in a post. In this work, we introduce the novel task of Claim Span Identification (CSI). We propose CURT, a large-scale Twitter corpus with token-level claim spans on more than 7.5k tweets. Furthermore, along with the standard token classification baselines, we benchmark our dataset with DABERTa, an adapter-based variation of RoBERTa. The experimental results attest that DABERTa outperforms the baseline systems across several evaluation metrics, improving by about 1.5 points. We also report detailed error analysis to validate the model's performance along with the ablation studies. Lastly, we release our comprehensive span annotation guidelines for public use.

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39. · 62% match · 2019 · 153 cit/yr

Understanding Conspiracy Theories (link)
Karen M. Douglas et al.
Political Psychology · Mar 20, 2019 · 1081 citations

Scholarly efforts to understand conspiracy theories have grown significantly in recent years, and there is now a broad and interdisciplinary literature that we review in this article. We ask three specific questions. First, what are the factors that are associated with conspiracy theorizing? Our review of the literature shows that conspiracy beliefs result from a range of psychological, political and social factors. Next, how are conspiracy theories communicated? Here, we explain how conspiracy theories are shared among individuals and spread through traditional and social media platforms. Next, what are the risks and rewards associated with conspiracy theories? By focusing on politics and science, we argue that conspiracy theories do more harm than good. Finally, because this is a growing literature and many open questions remain, we conclude by suggesting several promising avenues for future research.

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40. · 60% match · 2022 · 54 cit/yr

What Are Conspiracy Theories? A Definitional Approach to Their Correlates, Consequences, and Communication. (link)
Karen M. Douglas and Robbie M. Sutton
Annual review of psychology · Sep 28, 2022 · 190 citations

Conspiracy theories are abundant in social and political discourse, with serious consequences for individuals, groups, and societies. However, psychological scientists have started paying close attention to them only in the past 20 years. We review the spectacular progress that has since been made and some of the limitations of research so far, and we consider the prospects for further progress. To

this end, we take a step back to analyze the defining features that make conspiracy theories different in kind from other beliefs and different in degree from each other. We consider how these features determine the adoption, consequences, and transmission of belief in conspiracy theories, even though their role as causal or moderating variables has seldom been examined. We therefore advocate for a research agenda in the study of conspiracy theories that starts-as is routine in fields such as virology and toxicology-with a robust descriptive analysis of the ontology of the entity at its center. Expected final online publication date for the Annual Review of Psychology, Volume 74 is January 2023. Please see http://www.annualreviews.org/page/journal/pubdates for revised estimates.

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41. · 58% match · 2023 · 33 cit/yr

Conspiracy theories in online environments: An interdisciplinary literature review and agenda for future research (link)

Daniela Mahl, Mike S. Schäfer, and Jing Zeng

New Media & Society · Jul 1, 2023 · 91 citations

Research on conspiracy theories in digital media has grown considerably in recent years. As a result, the field of research has become more multidisciplinary and diverse. To bridge disciplinary boundaries, identify foci of analysis and research gaps, this study provides an interdisciplinary systematic literature review (2007–2020), analyzing current research on conspiracy theorizing online, both quantitatively and qualitatively. Findings show that the majority of studies lack a definition of conspiracy theories and fail to conceptually delineate conspiracy theories from other forms of deceptive content. We also found that while the field employs a variety of methodological approaches, most studies have focused on individual, "mainstream" social media platforms, "Western" countries, English-language communication, and single conspiracy theories. We use the findings of our review to remedy conceptual and empirical shortcomings and to provide suggestions on how to move forward in research on conspiracy theories online.

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42. · 56% match · 2019 · 2.7 cit/yr

Conspiratorial Narratives in Violent Political Actors' Language (link)

Stephane J. Baele

Journal of Language and Social Psychology · Aug 12, 2019 · 18 citations

This article articulates the concept of "conspiratorial narratives"—defined as stories which integrate a large range of events and archetypal characters from past and present in a single teleological explanation for the alleged suffering of a given social group—and argues that this particular linguistic construct is a key marker of extremist language. Using three different cases to illustrate our theoretical contribution (Nazi propaganda, Rwandan genocidaires' radio, IS' messaging), we show that paying attention to conspiratorial narratives leads us to significantly revise classic accounts of violent actors' language, and provides a better understanding of the link between that language and violence itself—more precisely, why violence happens, how much violence is directed to whom, and when it occurs.

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43. · 55% match · 2021 · 12 cit/yr

The language of conspiracy: A psychological analysis of speech used by conspiracy theorists and their followers on Twitter (link)

Amos Fong, J. Roozenbeek, Danielle Goldwert, Steven Rathje, and S. van der Linden

Group Processes & Intergroup Relations · May 31, 2021 · 59 citations

This paper analyzes key psychological themes in language used by prominent conspiracy theorists and science advocates on Twitter, as well as those of a random sample of their follower base. We conducted a variety of psycholinguistic analyses over a corpus of 16,290 influencer tweets and 160,949 follower tweets in order to evaluate stable intergroup differences in language use among those who subscribe or are exposed to conspiratorial content and those who are focused on scientific content. Our results indicate significant differences in the use of negative emotion (e.g., anger) between the two groups, as well as a focus, especially among conspiracy theorists, on topics such as death, religion, and power. Surprisingly, we found less pronounced differences in cognitive processes (e.g., certainty) and outgroup language. Our results add to a growing literature on the psychological characteristics underlying a "conspiracist worldview."

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44. · 52% match · 2021 · 9.2 cit/yr

Detection of conspiracy propagators using psycho-linguistic characteristics (link)

Anastasia Giachanou, Bilal Ghanem, and Paolo Rosso

Journal of Information Science · Jan 27, 2021 · 48 citations

The rise of social media has offered a fast and easy way for the propagation of conspiracy theories and other types of disinformation. Despite the research attention that has received, fake news detection remains an open problem and users keep sharing articles that contain false statements but which they consider real. In this article, we focus on the role of users in the propagation of conspiracy theories that is a specific type of disinformation. First, we compare profile and psycho-linguistic patterns of online users that tend to propagate posts that support conspiracy theories and of those who propagate posts that refute them. To this end, we perform a comparative analysis over various profile, psychological and linguistic characteristics using social media texts of users that share posts about conspiracy theories. Then, we compare the effectiveness of those characteristics for predicting whether a user is a conspiracy propagator or not. In addition, we propose ConspiDetector, a model that is based on a convolutional neural network (CNN) and which combines word embeddings with psycho-linguistic characteristics extracted from the tweets of users to detect conspiracy propagators. The results show that ConspiDetector can improve the performance in detecting conspiracy propagators by 8.82% compared with the CNN baseline with regard to F1-metric.

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45. · 50% match · 2019 · 15 cit/yr

Pathways to conspiracy: The social and linguistic precursors of involvement in Reddit's conspiracy theory forum (link)

C. Klein, Peter Clutton, and A. Dunn

PLoS ONE · Nov 18, 2019 · 98 citations

Many individuals who engage with conspiracy theories come to do so through a combination of individual and social factors. The interaction between these factors is challenging to study using traditional experimental designs. Reddit.com is a large connected set of online discussion forums, including one (r/conspiracy) devoted to wide-ranging discussion of conspiracy theories. The availability of large datasets of user comments from Reddit give a unique opportunity to observe human behavior in social spaces and at scale. Using a retrospective case control study design, we analyzed how Reddit users who would go on to engage with a conspiracy-related forum differed from other users in the language they use, differences in the social environments where they posted, and potential interactions between the two factors. Together, the analyses provide evidence for self-selection into communities with a shared set of interests which can feed into a conspiratorial world-view, and that these differences are detectable relative to controls even before users begin to post in r/conspiracy. We also suggest that survey-based and experimental studies may benefit from differentiating between passive private endorsement by individuals and active engagement with conspiracy theories in social spaces.

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46. · 48% match · 2022 · 4.5 cit/yr

Interconnectedness and (in)coherence as a signature of conspiracy worldviews (link)

A. Miani, T. Hills, and Adrian Bangerter

Science Advances · Oct 1, 2022 · 16 citations

Conspiracy theories may arise out of an overarching conspiracy worldview that identifies common elements of subterfuge across unrelated or even contradictory explanations, leading to networks of self-reinforcing beliefs. We test this conjecture by analyzing a large natural language database of conspiracy and nonconspiracy texts for the same events, thus linking theory-driven psychological research with data-driven computational approaches. We find that, relative to nonconspiracy texts, conspiracy texts are more interconnected, more topically heterogeneous, and more similar to one another, revealing lower cohesion within texts but higher cohesion between texts and providing strong empirical support for an overarching conspiracy worldview. Our results provide inroads for classification algorithms and further exploration into individual differences in belief structures.

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47. · 44% match · 2013 · 9.5 cit/yr

“What about building 7?” A social psychological study of online discussion of 9/11 conspiracy theories (link)

M. Wood and Karen M. Douglas

Frontiers in Psychology · Jul 8, 2013 · 121 citations

Recent research into the psychology of conspiracy belief has highlighted the importance of belief systems in the acceptance or rejection of conspiracy theories. We examined a large sample of conspiracist (pro-conspiracy-theory) and conventionalist (anti-conspiracy-theory) comments on news websites in order to investigate the relative importance of promoting alternative explanations vs. rejecting conventional explanations for events. In accordance with our hypotheses, we found that conspiracist commenters were more likely to argue against the opposing interpretation and less likely to argue in favor of their own interpretation, while the opposite was true of conventionalist commenters. However,

conspiracist comments were more likely to explicitly put forward an account than conventionalist comments were. In addition, conspiracists were more likely to express mistrust and made more positive and fewer negative references to other conspiracy theories. The data also indicate that conspiracists were largely unwilling to apply the "conspiracy theory" label to their own beliefs and objected when others did so, lending support to the long-held suggestion that conspiracy belief carries a social stigma. Finally, conventionalist arguments tended to have a more hostile tone. These tendencies in persuasive communication can be understood as a reflection of an underlying conspiracist worldview in which the details of individual conspiracy theories are less important than a generalized rejection of official explanations.

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48. · 42% match · 2018 · 16 cit/yr

Propagating and Debunking Conspiracy Theories on Twitter During the 2015–2016 Zika Virus Outbreak (link)

M. Wood

Cyberpsychology, Behavior and Social Networking · Aug 1, 2018 · 121 citations

Abstract The present study investigates the characteristics of discussion of conspiracy theories about the Zika virus outbreak of 2015–16 on Twitter. Content and social network analysis of a dataset of 25,162 original Tweets about Zika virus conspiracy theories showed that relative to debunking messages, conspiracy theories spread through a more decentralized network, are more likely to invoke supposedly knowledgeable authorities in making arguments, and ask more rhetorical questions. These trends can be understood in the context of previous work on conspiracy theories, including the "just asking questions" style of rhetoric, the importance of sourcing and authority, and the tendency to simultaneously consider many different potential conspiracies that might underlie an important topic or event.

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49. · 34% match · 2021 · 5.7 cit/yr

Human Rationales as Attribution Priors for Explainable Stance Detection (link)

Sahil Jayaram and Emily Allaway

Conference on Empirical Methods in Natural Language Processing · 30 citations

As NLP systems become better at detecting opinions and beliefs from text, it is important to ensure not only that models are accurate but also that they arrive at their predictions in ways that align with human reasoning. In this work, we present a method for imparting human-like rationalization to a stance detection model using crowdsourced annotations on a small fraction of the training data. We show that in a data-scarce setting, our approach can improve the reasoning of a state-of-the-art classifier—particularly for inputs containing challenging phenomena such as sarcasm—at no cost in predictive performance. Furthermore, we demonstrate that attention weights surpass a leading attribution method in providing faithful explanations of our model's predictions, thus serving as a computationally cheap and reliable source of attributions for our model.

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50. · 33% match · 2016 · 17 cit/yr

Rationale-Augmented Convolutional Neural Networks for Text Classification (link)

Ye Zhang, I. Marshall, and Byron C. Wallace

Proceedings of the Conference on Empirical Methods in Natural Language Processing. Conference on Empirical Methods in Natural Language Processing · May 14, 2016 · 171 citations

We present a new Convolutional Neural Network (CNN) model for text classification that jointly exploits labels on documents and their constituent sentences. Specifically, we consider scenarios in which annotators explicitly mark sentences (or snippets) that support their overall document categorization, i.e., they provide rationales. Our model exploits such supervision via a hierarchical approach in which each document is represented by a linear combination of the vector representations of its component sentences. We propose a sentence-level convolutional model that estimates the probability that a given sentence is a rationale, and we then scale the contribution of each sentence to the aggregate document representation in proportion to these estimates. Experiments on five classification datasets that have document labels and associated rationales demonstrate that our approach consistently outperforms strong baselines. Moreover, our model naturally provides explanations for its predictions.

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